

Assignment 06: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
library(tidyverse)
library(rvest)
library(here);
here()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
library(lubridate)

# Set theme
mytheme <- theme_classic() +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)

getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Set the year to 2024
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
lwsp_url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024"
webpage <- read_html(lwsp_url)
webpage
```

```
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
# selectors from SelectorGadget
water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
contact_person_tag    <- "div+ table tr:nth-child(4) td:nth-child(2)"
ownership_tag         <- "div+ table tr:nth-child(2) td:nth-child(4)"

# 12 monthly Max Day Use (MGD)
max_day_use_tag <- "th~ td+ td"

# scrape system-info fields
water_system_name <- webpage %>%
  html_nodes(water_system_name_tag) %>%
```

```

html_text() %>%
str_squish() %>%
`[`(1)

contact_person <- webpage %>%
  html_nodes(contact_person_tag) %>%
  html_text() %>%
  str_squish() %>%
  `[`(1)

ownership <- webpage %>%
  html_nodes(ownership_tag) %>%
  html_text() %>%
  str_squish() %>%
  `[`(1)

# scrape monthly max day use
max_day_use_chr <- webpage %>%
  html_nodes(max_day_use_tag) %>%
  html_text() %>%
  str_squish()

max_day_use_chr <- max_day_use_chr[str_detect(max_day_use_chr, "[0-9]+(\\. [0-9]+)?$")]
max_day_use_chr <- max_day_use_chr[1:12]

water_system_name

## [1] "Durham"

contact_person

## [1] "Reginald Hicks"

ownership

## [1] "Municipality"

max_day_use_chr

## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"

length(max_day_use_chr)

## [1] 12

```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

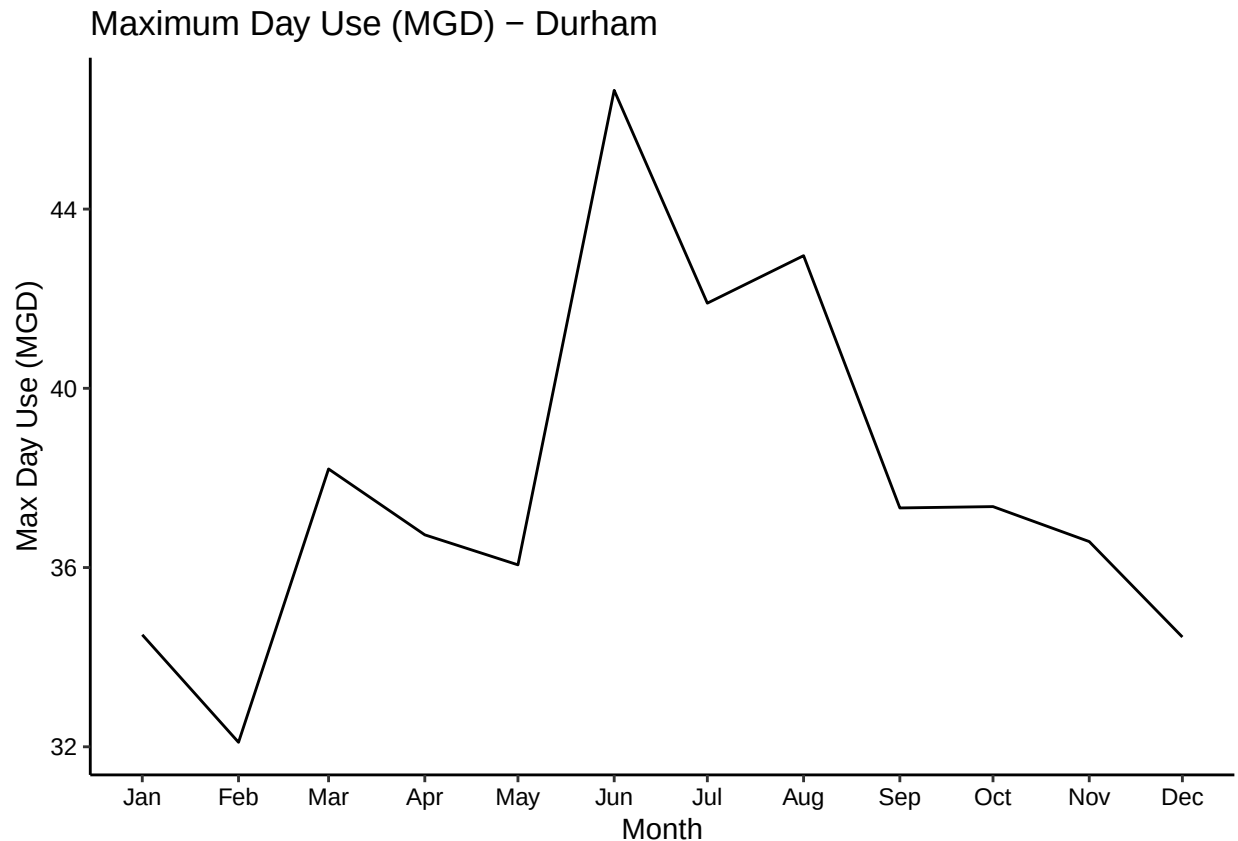
```
#4
#4 Create dataframe

df_2024 <- data.frame(
  Month = c(
    "Jan", "May", "Sep",
    "Feb", "Jun", "Oct",
    "Mar", "Jul", "Nov",
    "Apr", "Aug", "Dec"
  ),
  Year = rep(2024, 12),
  Water_System_Name = rep(water_system_name, 12),
  Contact_Person = rep(contact_person, 12),
  Ownership = rep(ownership, 12),
  Max_Day_Use_MGD = as.numeric(max_day_use_chr)
)

df_2024 <- df_2024 %>%
  mutate(
    Date = ymd(paste(Year, match(Month, month.abb), "01", sep = "-"))
  )

#5 Plot

ggplot(df_2024, aes(x = Date, y = Max_Day_Use_MGD)) +
  geom_line() +
  labs(
    title = paste("Maximum Day Use (MGD) -", water_system_name),
    x = "Month",
    y = "Max Day Use (MGD)"
  ) +
  scale_x_date(
    date_breaks = "1 month",
    date_labels = "%b"
  )
)
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6
lwsp_scrape <- function(PWSID, year) {

  the_url <- paste0(
    "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=",
    PWSID,
    "&year=",
    year
  )

  webpage <- read_html(the_url)

  water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  contact_person_tag    <- "div+ table tr:nth-child(4) td:nth-child(2)"
  ownership_tag         <- "div+ table tr:nth-child(2) td:nth-child(4)"
  max_day_use_tag       <- "th~ td+ td"
```

```

water_system_name <- webpage %>%
  html_nodes(water_system_name_tag) %>%
  html_text() %>%
  str_squish() %>%
  `[`(1)

contact_person <- webpage %>%
  html_nodes(contact_person_tag) %>%
  html_text() %>%
  str_squish() %>%
  `[`(1)

ownership <- webpage %>%
  html_nodes(ownership_tag) %>%
  html_text() %>%
  str_squish() %>%
  `[`(1)

max_day_use_chr <- webpage %>%
  html_nodes(max_day_use_tag) %>%
  html_text() %>%
  str_squish()

max_day_use_chr <- max_day_use_chr[str_detect(max_day_use_chr, "[0-9]+(\\.[0-9]+)?$")]
max_day_use_chr <- max_day_use_chr[1:12]

df <- data.frame(
  PWSID = PWSID,
  Month = c("Jan", "May", "Sep",
            "Feb", "Jun", "Oct",
            "Mar", "Jul", "Nov",
            "Apr", "Aug", "Dec"),
  Year = rep(year, 12),
  Water_System_Name = rep(water_system_name, 12),
  Contact_Person = rep(contact_person, 12),
  Ownership = rep(ownership, 12),
  Max_Day_Use_MGD = as.numeric(max_day_use_chr)
) %>%
  mutate(
    Date = ymd(paste(Year, match(Month, month.abb), "01", sep = "-"))
  )

return(df)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

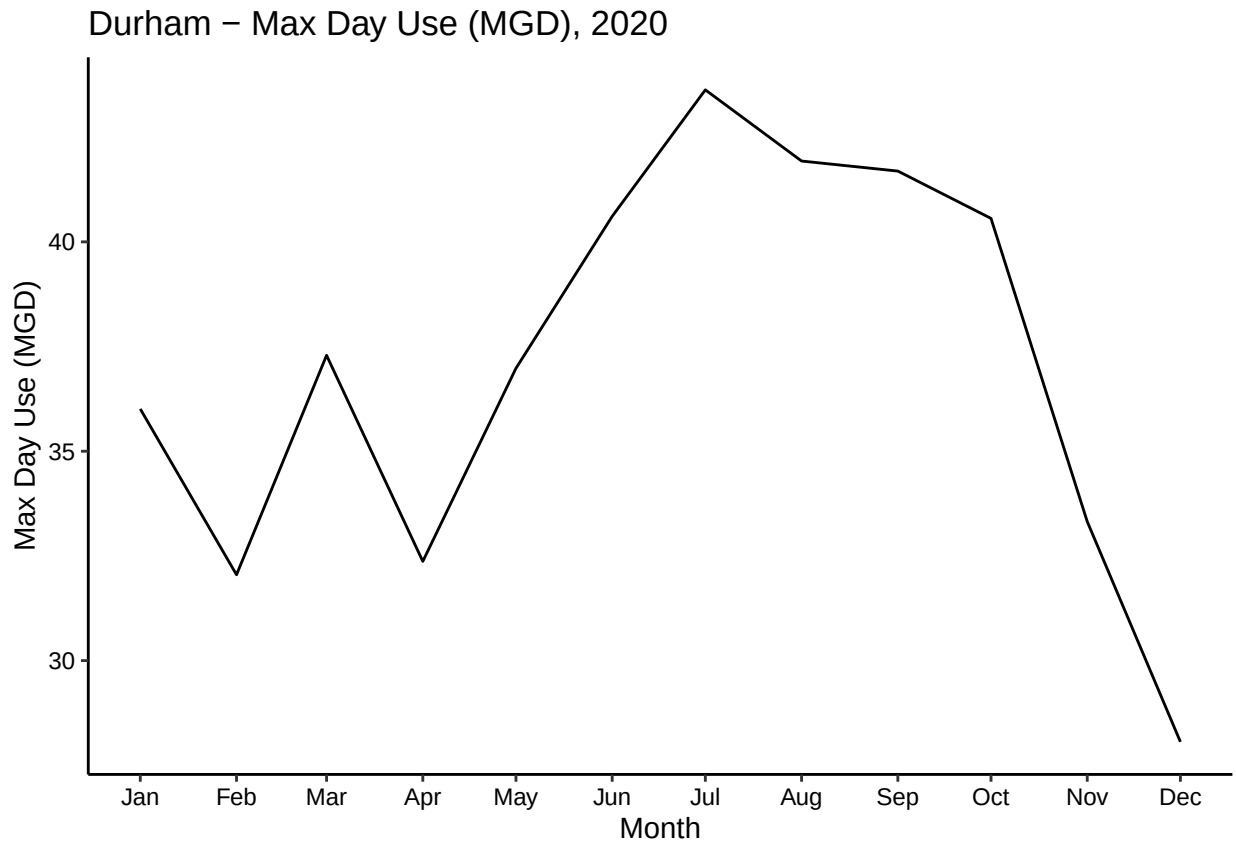
```

#7
durham_2020 <- lwsp_scrape("03-32-010", 2020)

ggplot(durham_2020, aes(x = Date, y = Max_Day_Use_MGD)) +

```

```
geom_line() +
labs(
  title = "Durham - Max Day Use (MGD), 2020",
  x = "Month",
  y = "Max Day Use (MGD)"
)+
scale_x_date(
  date_breaks = "1 month",
  date_labels = "%b"
)
```



```
glimpse(durham_2020)
```

```
## Rows: 12
## Columns: 8
## $ PWSID      <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ Month      <chr> "Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Ju~
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ Contact_Person <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hicks"~
## $ Ownership  <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Max_Day_Use_MGD <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
## $ Date       <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 202~
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data

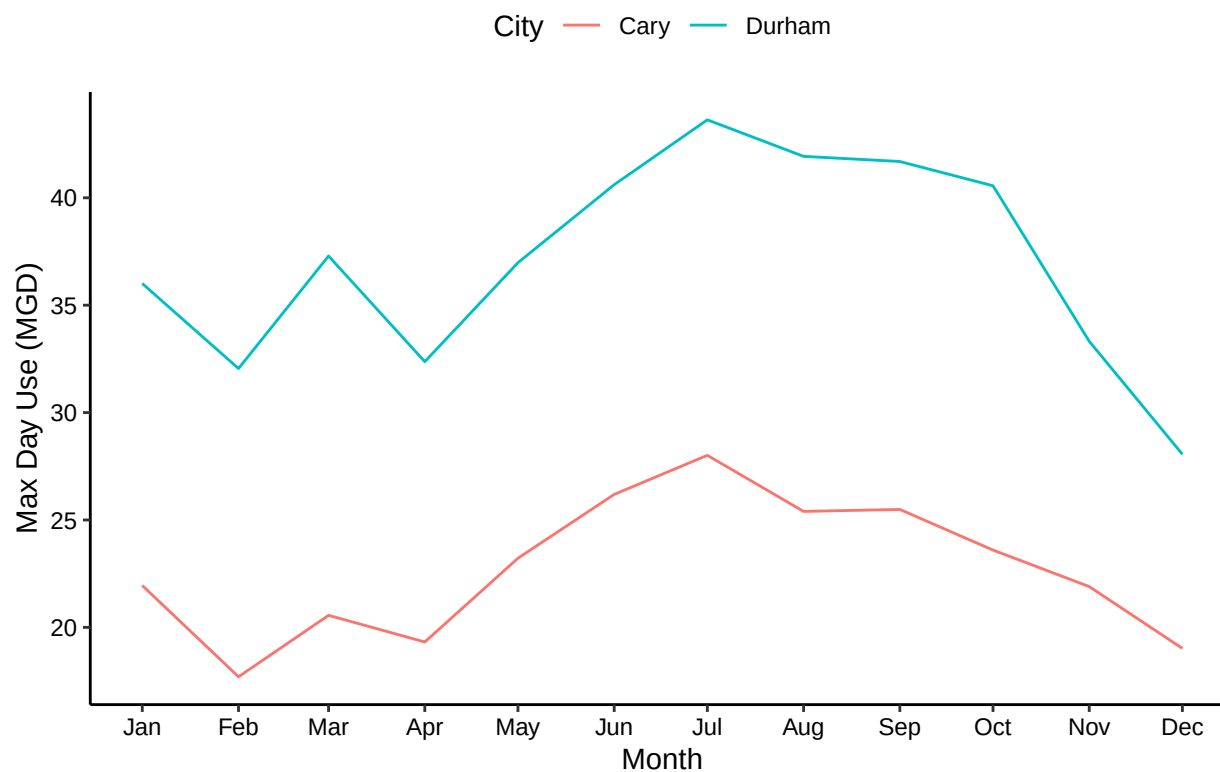
with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
cary_2020 <- lwsp_scrape("03-92-020", 2020)

compare_2020 <- bind_rows(
  durham_2020 %>% mutate(City = "Durham"),
  cary_2020 %>% mutate(City = "Cary")
)

ggplot(compare_2020, aes(x = Date, y = Max_Day_Use_MGD, color = City)) +
  geom_line() +
  labs(
    title = "Cary vs Durham - Max Day Use (MGD), 2020",
    x = "Month",
    y = "Max Day Use (MGD)"
  ) +
  scale_x_date(
    date_breaks = "1 month",
    date_labels = "%b"
  )
)
```

Cary vs Durham – Max Day Use (MGD), 2020



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the “06_Data_Scraping.Rmd” where we apply “map2()” to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
years <- 2018:2023
pwsids <- rep("03-92-020", length(years))

cary_multi <- map2(pwsids, years, lwsp_scrape) %>%
  bind_rows()

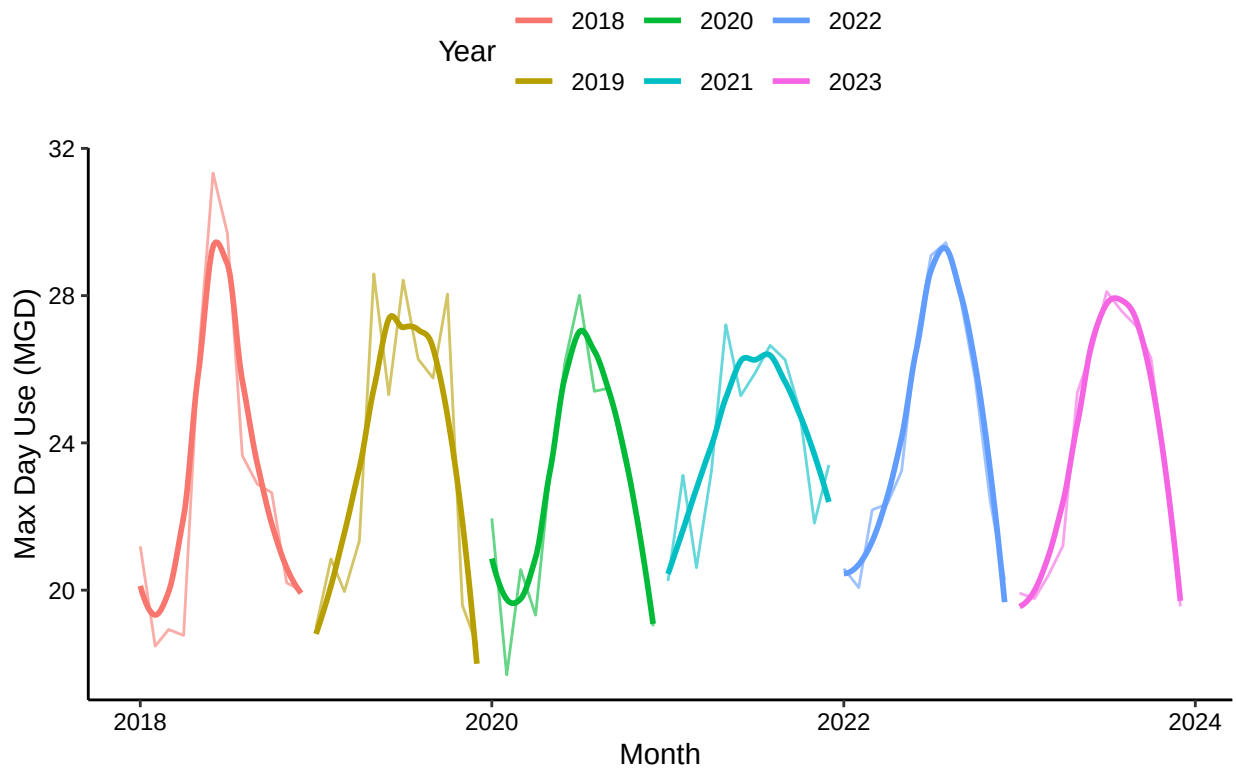
sum(is.na(cary_multi$Date))
```

```
## [1] 0
```

```
ggplot(cary_multi, aes(x = Date, y = Max_Day_Use_MGD, color = factor(Year))) +
  geom_line(alpha = 0.6) +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Cary Maximum Day Use (MGD), 2018-2023",
    x = "Month",
    y = "Max Day Use (MGD)",
    color = "Year"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Cary Maximum Day Use (MGD), 2018–2023



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: By looking at the plot, Cary does not appear to have a clear long-term upward or downward trend in water use from 2018 to 2023. The maximum daily withdrawals fluctuate from year to year, and there are noticeable seasonal patterns within each year. However, the overall level of water use seems relatively stable over time, without a consistent increase or decrease across the period.